

ADITHYA K

AI & Data Science Graduate | ML Engineer | Cloud AI Developer

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PROFESSIONAL SUMMARY

Motivated B.Tech graduate in Artificial Intelligence & Data Science (2025) with hands-on expertise in building production-grade GenAI systems, multi-agent pipelines, predictive ML models, and cloud-deployed applications. Experienced with LangGraph, LangChain, RAG architectures, AWS Bedrock, and Pinecone. Completed AWS re/Start and AWS Cloud Practitioner certifications. Seeking an ML Engineer or Data Science role to drive real-world AI impact.

EDUCATION

B.Tech — Artificial Intelligence & Data Science 2021 – 2025
Dhanalakshmi Srinivasan College of Engineering, Coimbatore, Tamil Nadu | **CGPA: 8.5**
Higher Secondary (Class XII) — Hindu Higher Secondary School, Ambur 82% | 2021
SSLC (Class X) — Hindu Higher Secondary School, Ambur 74% | 2019

TECHNICAL & SOFT SKILLS

- Programming Languages: Python (NumPy, Pandas, Matplotlib, Scikit-learn), SQL, JavaScript
- AI / ML Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, LangChain, LangGraph
- GenAI & Agentic AI: Multi-Agent Pipelines, RAG Architecture, Pinecone Vector DB, Prompt Engineering, AWS Bedrock, Gemini API, Llama-3
- Cloud & DevOps: AWS (EC2, S3, IAM, SageMaker, Bedrock), Docker (Basics), Git, GitHub, Vercel
- Data Science: EDA, Statistical Modelling, Feature Engineering, NLP, Computer Vision, Power BI
- Web & Databases: Next.js, Flask, Node.js, Express, Supabase, REST APIs
- Soft Skills: Analytical Thinking, Problem-Solving, Adaptability, Team Collaboration, Communication

PROJECTS

MediGuard — Multi-Agent Clinical Decision Support System 2024 – 2025

- Situation:** Built a production-grade multi-agent AI system to assist doctors in clinical decision-making by combining real-time RAG retrieval with LLM reasoning over medical literature.
- Task:** Designed and implemented the full agentic pipeline, vector database integration, and AWS cloud deployment.
- Action:** Orchestrated multiple AI agents using LangGraph; implemented RAG with Pinecone as vector store and AWS Bedrock (Claude) for LLM inference; integrated NLP-based evidence retrieval from clinical documents.
- Result:** Delivered context-aware clinical recommendations with accurate source citation; demonstrated end-to-end GenAI pipeline from ingestion to inference on AWS.

Tech Stack: Python, LangGraph, LangChain, Pinecone, AWS Bedrock, RAG, NLP, Multi-Agent AI

Health Sense Nexus — AI-Based Real-Time Health Monitoring System 2024 – 2025

Final Year Project

- Situation:** Developed a real-time health monitoring solution using AI and simulated IoT sensors to track patient vitals (heart rate, SpO2, temperature) and detect anomalies automatically.

- **Task:** Responsible for data preprocessing, model training, REST API development, and cloud deployment on AWS.
- **Action:** Built LSTM neural networks using TensorFlow/Keras for anomaly prediction; deployed a Flask API on AWS EC2 and designed a Power BI dashboard for live visualisation.
- **Result:** ~94% anomaly detection accuracy; API handles ~500 predictions/hour with <200 ms response time; dashboard reduced manual review time by 60%.

Tech Stack: *Python, TensorFlow/Keras, LSTM, Flask, AWS (EC2, S3, IAM), MQTT, Power BI, Pandas, NumPy*

LinkedIn Job Search Automation — Python Automation Pipeline

2024

- **Situation:** Designed an automated job application and profile tracking pipeline to accelerate the job search process at scale.
- **Action:** Built with Python and APScheduler to automate scheduled job searches, track application status, and generate strategy outputs; integrated profile completeness scoring to optimise LinkedIn visibility.
- **Result:** Automated high-volume job applications pipeline; demonstrated applied Python automation and scheduling in a real-world career workflow.

Tech Stack: *Python, APScheduler, LinkedIn API, JSON, Automation Scripting*

INTERNSHIPS & CERTIFICATIONS

- **AWS Cloud Practitioner — Amazon Web Services (2025)**
- AWS re/Start Programme — Cultus & Amazon Web Services (2025)
- Data Science Job Simulation — BCG (Boston Consulting Group) (2024)
- Prompt Engineering for Generative AI — Internshala Trainings (2024)
- Networking Essentials — Cisco Networking Academy (2023)
- Entry Level Python Programmer — Cisco Networking Academy (2023)

ACHIEVEMENTS

- Built end-to-end GenAI production pipeline (MediGuard) using LangGraph, AWS Bedrock, and Pinecone RAG — covering multi-agent orchestration, vector retrieval, and cloud inference.
- Achieved ~94% anomaly detection accuracy in final year IoT health monitoring project deployed on AWS EC2.
- Completed AWS re/Start Programme — a competitive, instructor-led training certified by Amazon Web Services.
- Completed 50+ hours of hands-on AI/ML labs covering the full ML lifecycle from data ingestion to deployment.

LANGUAGES

English (Professional) • **Tamil** (Native)